SVI and SVI-JW

Stochastic-volatility-inspired slices, jump-wing handles, and constrained calibration

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Abstract

This note documents the Vol-Fitter's SVI slice model. Raw SVI parametrizes total implied variance as a hyperbola in log-moneyness, $w(k) = a + b\{\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\}$, with five parameters that are geometrically opaque but analytically convenient. The *jump-wing* (SVI-JW) re-coordinatization replaces them with five quantities a trader actually quotes — ATM variance, ATM skew, the two wing slopes, and the minimum variance — and we give the exact conversion in both directions. We state the static no-arbitrage conditions (Durrleman's butterfly function and Lee's wing bound), show how the calibrator enforces them as two soft penalties that vanish on any admissible slice, and explain the unconstrained reparametrization (softplus / tanh / exp) that lets a plain Levenberg–Marquardt solver only ever propose feasible slices. As in the LQD note, the residual Jacobian is available in closed form; a fresh run recovers a benchmark SVI-JW smile to machine precision, reproduces its jump-wing handles exactly, and runs 2.41× faster than the finite-difference Jacobian.

Contents

1 Why SVI, and why two parametrizations

SVI is the workhorse of single-slice equity-index fitting because the hyperbola

$$w(k) = a + b\Big\{\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\Big\} \tag{1}$$

is, in five parameters, simultaneously (i) linear in its wings — $w(k) \sim b(1 \pm \rho)\,|k|$ — so automatically Lee-compatible, (ii) smooth and convex enough to be butterfly-free over a wide admissible cone, and (iii) cheap to evaluate and differentiate. The cost is that (a, b, ρ, m, σ) are geometrically entangled: no single one is "the skew" or "the ATM level". The Vol-Fitter therefore carries SVI in *two* coordinate systems — raw for the optimizer, jump-wing for the human — and converts losslessly between them.

Heuristic

Read (1) as a hyperbola: b is the overall steepness (the asymptotic opening angle), $\rho \in (-1, 1)$ tilts it (left wing steeper than right for the equity put skew $\rho < 0$), m shifts its vertex in k, σ rounds off the vertex (larger σ = flatter bottom), and a raises the whole curve. The wings are straight lines of slope $b(1-\rho)$ on the left and $b(1+\rho)$ on the right — which is exactly what Lee's bound constrains.

2 Raw SVI geometry

The derivatives used everywhere below are

$$w'(k) = b\Big(\rho + \frac{k-m}{r}\Big), \qquad w''(k) = \frac{b\,\sigma^2}{r^3}, \qquad r = \sqrt{(k-m)^2 + \sigma^2}. \tag{2}$$

Note $w'' > 0$ always (the slice is convex in k), and the wing limits are

$$\lim_{k\to-\infty} \frac{w(k)}{|k|} = b(1-\rho), \qquad \lim_{k\to+\infty} \frac{w(k)}{k} = b(1+\rho). \tag{3}$$

The vertex (minimum total variance) sits at $k^\star = m - \sigma\rho/\sqrt{1-\rho^2}$ with value $w(k^\star) = a + b\sigma\sqrt{1-\rho^2}$; this minimum is the quantity the jump-wing parametrization calls $\widetilde{v}\,t$.

3 The jump-wing handles

The SVI-JW parametrization $(v, \psi, p, c, \widetilde{v})$ at expiry t is defined by ATM and wing functionals of the slice:

$$\begin{aligned}
v &= \frac{w(0)}{t} && \text{ATM variance,} \\
\psi &= \frac{b}{2\sqrt{w_0}}\Big(\rho - \frac{m}{\sqrt{m^2+\sigma^2}}\Big) && \text{ATM skew,} \\
p &= \frac{b(1-\rho)}{\sqrt{w_0}}, \quad c = \frac{b(1+\rho)}{\sqrt{w_0}} && \text{put/call wing slopes,} \\
\widetilde{v} &= \frac{a + b\sigma\sqrt{1-\rho^2}}{t} && \text{minimum variance,}
\end{aligned} \tag{4}$$

with $w_0 = w(0) = vt$. Figure 1 annotates the five handles on a fitted smile.

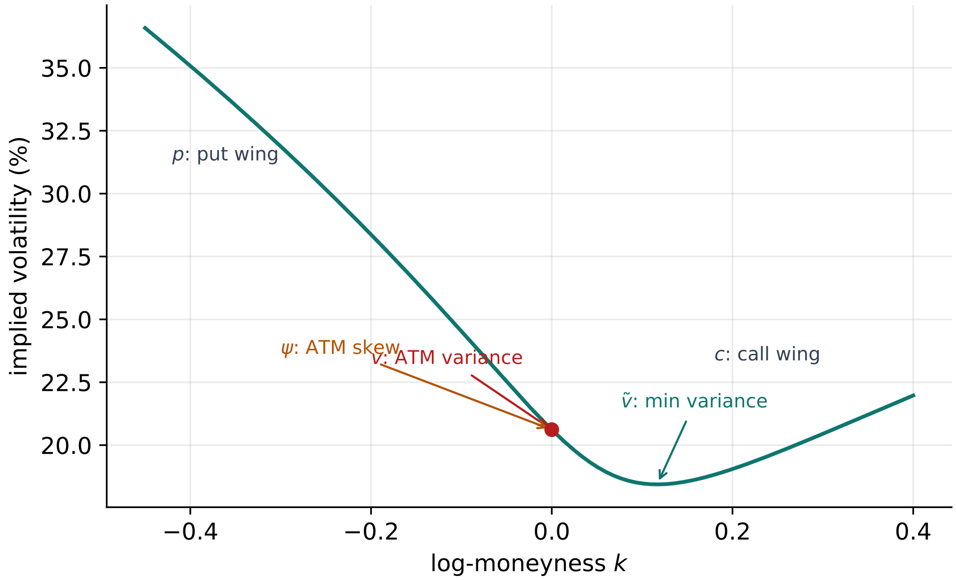


Figure 1: The five SVI-JW handles read off a fitted slice: ATM variance v, ATM skew ψ, put/call wing slopes p, c, and minimum variance $\widetilde{v}$.

3.1 The conversion, both ways

Going *from* jump-wing *to* raw is what the app stores when a desk quotes handles. Set $w_0 = vt$, $\sqrt{w_0} = \sqrt{vt}$, and

$$b = \frac{\sqrt{w_0}}{2}(p + c), \qquad \rho = \frac{c - p}{c + p}, \qquad \chi := \frac{m}{\sqrt{m^2 + \sigma^2}} = \rho - \frac{4\psi}{p + c}. \tag{5}$$

From χ one recovers $m = \chi\sigma/\sqrt{1 - \chi^2}$ and $\sqrt{m^2 + \sigma^2} = \sigma/\sqrt{1 - \chi^2}$; subtracting the minimum-variance identity from the ATM identity gives the closed form

$$\sigma = \frac{w_0 - \widetilde{v}\, t}{b\big((1 - \rho\chi)/\sqrt{1 - \chi^2} - \sqrt{1 - \rho^2}\big)}, \qquad a = \widetilde{v}\, t - b\sigma\sqrt{1 - \rho^2}. \tag{6}$$

The reverse map *from* raw *to* jump-wing is just the definitions (4) evaluated on the fitted raw slice; the Vol-Fitter uses it to display handles after every calibration. Section 6 verifies the round-trip numerically to machine precision.

4 Static no-arbitrage

4.1 Butterfly: Durrleman's function

A slice is free of butterfly arbitrage iff the risk-neutral density it implies is non-negative, which Durrleman's condition expresses directly on total variance:

$$g(k) = \Big(1 - \frac{k\,w'(k)}{2w(k)}\Big)^2 - \frac{w'(k)^2}{4}\Big(\frac{1}{w(k)} + \frac{1}{4}\Big) + \frac{w''(k)}{2} \;\geq\; 0 \quad \forall k. \tag{7}$$

For raw SVI, g is a smooth function of (1)–(2) with no closed-form sign, so it is checked on a grid as a fit diagnostic. The Gatheral–Jacquier conditions give sufficient parameter-space inequalities bounding the wing steepness $b(1+|\rho|)$ and requiring the minimum total variance to be non-negative; the calibrator enforces the two that bite in practice.

4.2 The two soft penalties

The optimizer adds, to the data residual, two penalties that are *exactly zero* on an admissible slice and grow linearly past the boundary:

$$\rho_{\text{pen}}(\theta) = W \begin{pmatrix} \max\big(-(a + b\sigma\sqrt{1-\rho^2}),\, 0\big) \\ \max\big(b(1+|\rho|) - \beta_{\max},\, 0\big) \end{pmatrix}, \tag{8}$$

with weight $W = 1000$ and Lee cap $\beta_{\max} = 2.0$. The first keeps the *minimum total variance* non-negative (no negative density mass at the vertex); the second enforces Lee's bound that the asymptotic total-variance slope cannot exceed 2. Because both vanish on a clean fit, an admissible smile (such as the benchmark of section 6) is byte-identical with or without them — they are a feasibility fence, not a regularizer.

Caution

The penalty weight $W = 1000$ is in vol^2 units, deliberately large enough to dominate a violated constraint yet invisible on an admissible slice. It is *not* a tuning dial for fit quality: raising it cannot improve a feasible fit, and lowering it only lets the optimizer wander into the arbitrageable cone. Note 09 develops the Lee bound across all the models; here it is one of these two hinges.

5 Calibration

5.1 Unconstrained reparametrization

Rather than run a bound-constrained optimizer, the Vol-Fitter reparametrizes so that *every* point of $\mathbb{R}^5$ maps to an admissible raw slice:

$$a \text{ free}, \quad b = \text{softplus}(\theta_b) \geq 0, \quad \rho = \tanh(\theta_\rho) \in (-1,1), \quad m \text{ free}, \quad \sigma = e^{\theta_\sigma} > 0. \tag{9}$$

This guarantees $b \geq 0$, $|\rho| < 1$, $\sigma > 0$ for free, so the unconstrained Levenberg–Marquardt solver never wastes effort on the box and never proposes a nonsensical hyperbola; only the two *economic* constraints (8) remain, and they are soft.

5.2 Objective, weights and the data-driven start

The data term is a plain implied-volatility residual, $r_i = \sqrt{\omega_i}\,(\sigma^{\mathrm{SVI}}(k_i) - \sigma_i)$, optionally vega-weighted (pass $\omega_i = \mathrm{vega}_i^2$). The initializer is geometric and reads the smile directly: the argmin of w seeds (a, m), and the two finite wing slopes seed (b, ρ) via (3), then the reparametrization (9) is inverted to get θ_0. On a liquid smile a single LM pass converges from this start.

5.3 The analytic Jacobian and the optional blocks

The residual stack — data (mid or band), the two penalties, and an optional calendar hinge — has a closed-form Jacobian propagated through the reparametrization (9) by the chain rule; the calibrator uses it whenever no var-swap or prior block is present (those are not yet differentiated and fall back to finite differences, exactly as in LQD). The optional blocks mirror the rest of the series:

- **Band fit** (Note 07): the mid residual becomes a vol-space band hinge with a small mid anchor.
- **Calendar hinge** (Note 10): the *model-agnostic* constraint $\sqrt{W_{\mathrm{cal}}}\,\max(\mathrm{floor} - w(k), 0)$ keeps this slice's total variance at or above the nearer expiry's — the same machinery LQD expresses via its quantile $A(z)$, here written directly on $w(k)$.
- **Var-swap target** (Note 08) and **prior persistence** (Note 13): added as in LQD, so the SVI *display overlay* receives the same prior treatment as the primary model — closing a long-standing asymmetry in which SVI overlays got no prior at all.

Performance

The solver is Levenberg–Marquardt (`method="lm"`), not trust-region reflective. On noisy real chains LM crosses the penalty kinks in far fewer iterations than `trf` at matched tolerance, so it is the faster same-accuracy choice; the analytic Jacobian is then a drop-in that swaps scipy's $1 + P$ finite-difference evaluations for one closed-form call. Section 6 measures $2.41\times$ end-to-end with the optimum unchanged to $3.9e-34$ in cost. See appendix B.

6 Worked example: round-trip recovery

The cleanest test of an SVI-JW implementation is a round trip. Take the SPX-like benchmark raw slice $(a, b, \rho, m, \sigma) = (0.010625,\ 0.07289,\ -0.5,\ 0.05831,\ 0.10100)$ at $t = 0.5$ — the image under (5)–(6) of the jump-wing handles $(v, \psi, p, c, \widetilde{v}) = (0.0425, -0.25, 0.75, 0.25, 0.034)$. Generate quotes from it, refit raw SVI from the geometric start, and read the handles back.

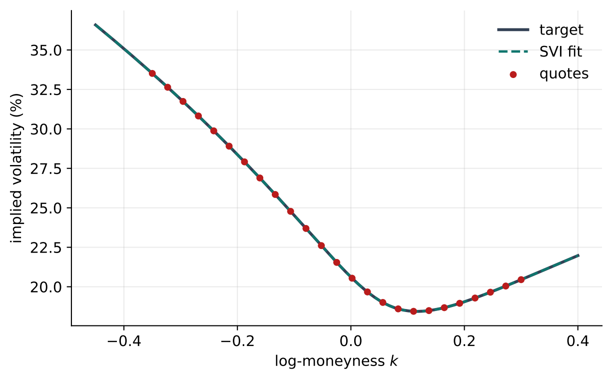
The fit recovers the raw parameters (table 1) with a maximum implied-volatility error of 0.00 volatility basis points in 21 residual evaluations, the wing slopes come out $b(1-\rho) = 0.1093$ (put) and $b(1+\rho) = 0.0364$ (call), and — the point of the exercise — the recovered jump-wing handles (table 2) reproduce the original $(0.0425, -0.25, 0.75, 0.25, 0.034)$ to display precision. The conversion (4)–(6) is therefore exact, not approximate. Figure 2 shows the fit and section 6 the Durrleman function staying non-negative across the strike range.

Table 1: Recovered raw SVI.

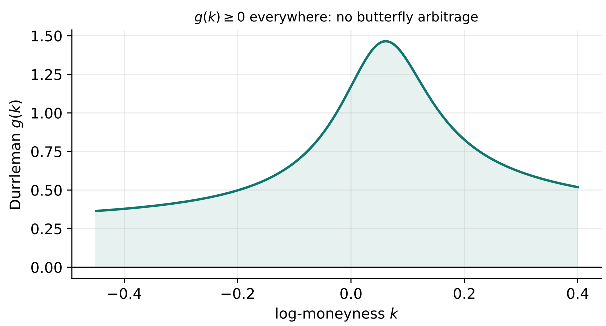
Parameter	Value
a	+0.010625
b	+0.072887
ρ	-0.500000
m	+0.058310
σ	+0.100995

Table 2: Recovered SVI-JW handles.

Handle	Value
v	+0.042500
ψ	-0.250000
p	+0.750000
c	+0.250000
$\tilde{v}$	+0.034000



(a) target, fit, quotes



(b) Durrleman $g(k) \geq 0$

Figure 2: Left: the benchmark smile and its SVI fit (max error 0.00 vol bps). Right: the butterfly diagnostic $g(k)$, positive throughout.

Heuristic

A round-trip error of 0.00 vol bps is expected because the target *is* an SVI slice — the exercise validates the optimizer and the conversion, not the expressiveness of SVI. On genuine market chains SVI typically fits liquid index smiles to a few vol bps but is stiffer than LQD or Local-Vol on event or double-hump shapes (a single hyperbola has one vertex), which is precisely why the app offers the other models.

7 What is original here, and limitations

SVI and the jump-wing handles are due to Gatheral; the arbitrage-free analysis is Gatheral–Jacquier and Durrleman. The implementation choices worth noting are the *feasibility-by-reparametrization* (9) (the solver cannot leave the admissible cone except through the two soft economic constraints), the *closed-form raw↔JW round trip* used live for trader handles, and the *analytic LM Jacobian* that makes SVI the speed peer of the other models. As a slice model SVI is intrinsically unimodal: it cannot represent a bimodal event density (Note 01's double-hat) or a WW-shaped smile (Note 03), and its single global hyperbola can over-smooth a sharp short-dated skew. It is butterfly-free only over its admissible cone — enforced softly here — and carries no calendar coupling on its own; that is supplied by the model-agnostic hinge of Note 10.

A Hyperparameter atlas

Table 3: SVI hyperparameters.

Knob	Default	Role
Surfaced (FitSettings)		
`sviPenaltyWeight` W	1000	Weight of the two soft no-arbitrage penalties (8); vol^2 units.
`leeSlopeMax` $\beta_{\max}$	2.0	Lee wing-slope cap; the asymptotic total-variance slope $b(1+\lvert\rho\rvert)$ may not exceed it.
`midAnchorWeight`	0.05	Mid anchor weight in the band-fit objective (Note 07).
`weightScheme`	`equal`	Per-quote weights (Note 07); pass vega^2 for a vega-weighted fit.
`calendarWeight`	10^6	Calendar hinge penalty (Note 10).
Internal (reparametrization / solver)		
$b = \mathrm{softplus}(\theta_b)$	—	Enforces $b \geq 0$.
$\rho = \tanh(\theta_\rho)$	—	Enforces $\lvert\rho\rvert < 1$.
$\sigma = e^{\theta_\sigma}$	—	Enforces $\sigma > 0$.
LM tolerances	10^{-15}	`xtol/ftol/gtol`; LM converges in few iterations so tight tolerances are cheap.
floor 10^{-12}	—	Total-variance floor inside $\sqrt{w/t}$ to avoid $\sqrt{\mathrm{neg}}$ on infeasible trial slices.

B Performance notes

1. **Levenberg–Marquardt over trust-region.** On real noisy chains LM crosses the penalty kinks in fewer iterations than `trf` at matched tolerance; `trf` at 10^{-10} was measured *slower* on real nodes.

2. **Analytic Jacobian.** Closed-form Jacobian of the residual stack (data + penalties + calendar) through the reparametrization; a fresh run measured 1.1 ms vs 2.7 ms for the finite-difference path (2.41×) with the optimum unchanged to $3.9e-34$ in cost. Gated to the var-swap/prior-free configuration.

3. **Feasibility by construction.** The reparametrization removes the box-constraint handling entirely, so each LM step is a plain linear solve.

4. **Geometric warm start.** Reading (a, m) from the vertex and (b, ρ) from the wing slopes lands θ_0 close enough that liquid smiles converge in a single pass.

References

[1] J. Gatheral. A parsimonious arbitrage-free implied volatility parameterization. Presentation, Global Derivatives, 2004.

[2] J. Gatheral and A. Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.

[3] R. W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004.

[4] V. Durrleman. From implied to spot volatilities. *Finance and Stochastics*, 14(2):157–177, 2010.